CHANDU PRASAD BHAIRAPU



AI Engineer

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ABOUT ME

Master's graduate in Artificial Intelligence with 3 years of professional experience in AI engineering and software development. Specialized in NLP with transformer models and LLMs, from information extraction on domain documents and fine-tuning with Hugging Face to evaluation pipelines and containerized deployment. Most recently sole developer of an agentic knowledge-extraction system for the German energy sector, backed by a strong foundation in machine learning on real industrial data and Python development across enterprise and start-up environments.

EXPERIENCE

Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

Kniit GmbH

November 2025 – present | Muldestausee, Germany

- Sole developer owning an NLP information-extraction pipeline end-to-end, from data ingestion and chunking to prompt engineering and indexing of complex regulatory and technical documents, reporting directly to the CTO.
- Built multi-agent LLM extraction (LangGraph: entity-extraction, relationship-analysis, consolidation and reflection agents in parallel) and evaluated it against an SAP-system-derived ground truth, raising precision/recall from 33%/45% to 93%/90%.
- Built a deterministic, code-based pipeline for a second, high-volume source stream, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing accuracy toward 100%.
- Designed a Model Context Protocol (MCP) service exposing the extracted knowledge graph with millisecond-latency retrieval, and built Expert Interface, a web app (React, Python microservices) on Azure where domain experts validate extractions.
- **Tech stack: Python, LLMs, LangGraph, LangChain, LiteLLM, MCP, FastAPI, Docker, Kubernetes, Azure, Neo4j, MongoDB, Postgres**

GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 | Leverkusen, Germany

- Designed and deployed an agentic AI application with LangGraph that routes requests to vector database search, general LLM inference or summarization; worked with business stakeholders on their requirements.
- Python/FastAPI backend and React front-end, containerized with Docker, deployed to AWS with CDK; versioned in GitLab with CI/CD.
- **Tech stack: Python, LangGraph, LangChain, FastAPI, Docker, AWS (CDK, EC2, Lambda), GitLab CI/CD, React**

GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 | Leverkusen, Germany

- Benchmarked multilingual transformer embedding models on over 1M chunks of internal chemical data and fine-tuned models for retrieval.
- Built an automated evaluation pipeline for the PrismaChat RAG assistant with RAGAS and PytrecEval against chemist-provided ground truth, running on AWS (Lambda, SQS, DynamoDB) deployed with CDK.
- Designed RAG pipelines: chunking strategies, vector databases and indexing.
- **Tech stack: Python, PyTorch, Hugging Face Transformers, Scikit-Learn, TensorFlow, LangChain, RAGAS, PytrecEval, AWS CDK, Docker, Azure AI Search**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Trained and compared Random Forest, XGBoost and LSTM models on time-series process data (lag features, rolling statistical windows; raw sequences for the LSTM) and benchmarked them against empirical curve-fitting.
- Delivered a Streamlit dashboard so chemists could simulate parameter changes and see predictions before running physical batches.
- **Tech stack: Python, Scikit-Learn, XGBoost, TensorFlow/Keras, Pandas, NumPy, SciPy, Streamlit**

Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XGBoost regressor, FB Prophet and LSTM models for EV load forecasting; the LSTM reached 96% test accuracy on unseen data.

PROJECTS

Project: Care Note Structurer | PyTorch | Hugging Face | MLflow | FastAPI | Docker

- Clinical NLP pipeline that turns dictated care-note transcripts into structured entries (vitals, medication, wound notes): a German encoder fine-tuned with Hugging Face Transformers for entity extraction, compared against LLM extraction with schema validation and LLM fallback.
- Experiment tracking and model versioning in MLflow, field-level evaluation on a labelled test set in CI, FastAPI service in Docker with latency logging and a pseudonymisation step before training.

Project: Meme Generation using LLMs | Open Llama 3B | QLoRA

- Fine-tuned an Open Llama 3B model with QLoRA to generate responses to educational meme-related prompts.

TECHNICAL SKILLSET

- **NLP & LLMs:** Hugging Face Transformers, transformer embedding models, LLMs, fine-tuning (QLoRA), information extraction, NLTK, RAG, Prompt Engineering, LangGraph, LangChain, LiteLLM
- **Machine Learning:** Python, PyTorch, TensorFlow/Keras, Scikit-Learn, XGBoost, Pandas, NumPy, evaluation (RAGAS, PytrecEval, precision/recall)
- **MLOps & Software Engineering:** Docker, Kubernetes, Git, GitLab CI/CD, FastAPI, AWS (CDK, Lambda, SQS, S3), Azure, Streamlit
- **Data:** Postgres, MongoDB, Neo4j, Vector Databases (Qdrant, Pinecone, Azure AI Search)

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Minor in Computer Science & Engineering — July 2019 – July 2021
G Pulla Reddy Engineering College, Kurnool, India

Bachelor of Technology in Mechanical Engineering — August 2017 – July 2021
G Pulla Reddy Engineering College, Kurnool, India

PUBLICATIONS & CERTIFICATIONS

- **[1]** Navigating the Future: An approach of Autonomous Indoor Vehicles
- Udemy - LLMs Mastery : Transformers & Generative AI
- Udemy - The Complete Neural Networks Boot camp

LANGUAGES

English (Business Fluent) | German (B1) | Hindi (Fluent) | Telugu (Native)